NAME: _____ Geometry (FL B.E.S.T.)

3.2 Reflections

A reflection flips a figure over a mirror line so that every point and its image are the same distance from the line, on opposite sides — which is why reflections preserve distance but reverse orientation, unlike translations. Beyond the four memorized rules (x-axis, y-axis, $y = x$, $y = -x$), this section covers reflecting over any horizontal or vertical line using a distance formula, proving algebraically why distance survives while orientation flips, and working backward to find the mirror line itself from a point and its image.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Reflect $(4, -2)$ over x-axis.

2. Reflect $(-3, 6)$ over y-axis.

3. Reflect $(5, 1)$ over $y = x$.

4. Reflect $(2, 3)$ over $y = -x$.

5. Reflect $(0, 4)$ over $x = 1$.

6. Reflect $(3, 0)$ over $y = 2$.

7. Reflection reverses orientation?

8. Figure congruent to its reflection?

9. Reflect $(-1, -8)$ over x-axis.

10. Reflect $(7, -3)$ over y-axis.

11. Segment joining a point to its image is bisected by what?

12. Reflecting over $x = 0$ is reflecting over which axis?

13. Reflecting over $y = 0$ is reflecting over which axis?

14. Triangle $P(1, 1)$, $Q(4, 1)$, $R(1, 5)$ is reflected over the y-axis. Find all three image vertices.

15. A point $(5, -2)$ reflects to $(5, 8)$. Find the equation of the line of reflection.

✓ Answer Key

1. $(4, 2)$
2. $(3, 6)$
3. $(1, 5)$
4. $(-3, -2)$
5. $(2, 4)$
6. $(3, 4)$
7. Yes
8. Yes
9. $(-1, 8)$
10. $(-7, -3)$
11. The reflection line
12. y-axis
13. x-axis
14. $P'(-1, 1), Q'(-4, 1), R'(-1, 5)$
15. $y = 3$